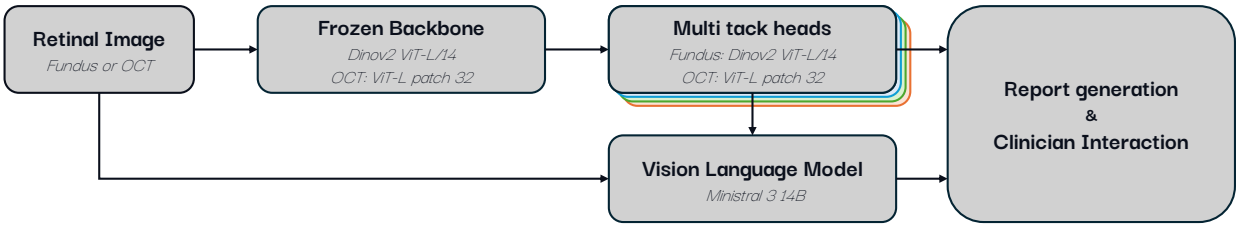
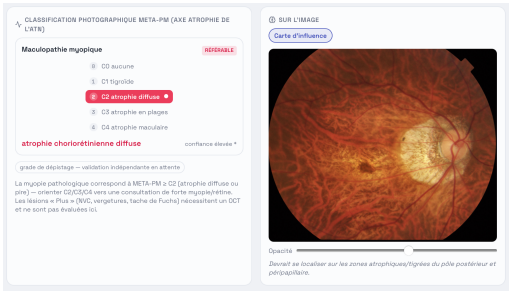
Deep learning for medical imaging: self-supervised backbone and multi-modal layer

Co-founder and deep learning engineer • Rodopsia • 2024–2026 • Python, PyTorch

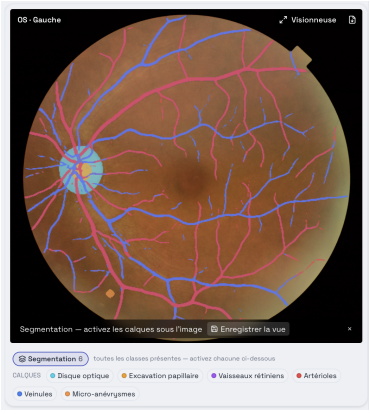
Rodopsia is a diagnostic-support platform for ophthalmic imaging (fundus and OCT) that I designed and built end to end. At its core is a DINOv2-based self-supervised backbone pretrained on large-scale unlabelled images, feeding specialised task heads (segmentation, image description and disease detection). On top sits a multimodal layer - a vision-language model - that combines image features with explanations, for a better interpretation of how the model reaches its choice. I own the whole chain: data pipeline, pretraining, evaluation, inference and the application itself. This is end-to-end deep-learning delivery: training and benchmarking state-of-the-art models, calibrating their outputs, and shipping them inside a product.



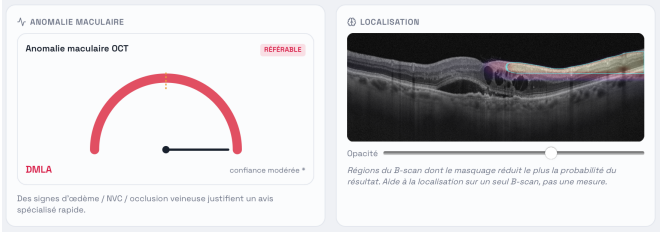
The Rodopsia architecture: a shared self-supervised backbone feeding specialised task heads, with a multimodal vision-language layer for explanation.



Example of a disease head for fundus



Example of anatomical and lesion segmentation for fundus



Example of a disease head for OCT, with a heatmap indicating where the model based its choice